

WNW Quantum Weft

Spherical narrow-web layers for estimate-first facts, small-device relay, P2P RAG slices,
and future quantum circuit development.

Positioning date: 2026-06-30. The ScienceDirect PII page supplied with the request was reachable only as a generic ScienceDirect shell in this environment, so its title is not restated. The design below uses it as directional spherical positioning and grounds claims in accessible sources listed at the end.

1. Design Thesis

WNW should become the efficient narrow-web layer between worlds and between small devices. It should not behave like a broad search engine by default. A user specifies information; WNW returns a proposal with result size and table shape before fetching complete facts.

- Spherical fabric over square grid: use layer, theta, phi, beat, band, and relation digest.
- Pulse is the vessel: it carries event timing within layers and between layers.
- Narrow web first: organized facts beat open-web search for specified questions.
- LLM second: call Google or an LLM only when the request is under-specified or a field is missing.
- Quantum Forge: code, circuit manifests, and MLflow-like run records share the same signed fabric.
- Vank Mesh Exchange: market UX without exchange custody or a central orderbook trust root.
- Always visual: every architecture surface needs a traversable AR/VR-ready demo.

2. Quantum Weft Layers

Layer	Name	Function
0	Pulse vessel	Clock, event root, beat envelope, and cross-layer event timing.
1	Quantum-dot anchor	Optical or indirect address surface: laser path, dot/molecule identity, emission band.
2	Bonded molecule ring	Element addresses bind through molecular bonds and neighbor digests.
3	Frequency shell	Outer layers may use different bands while staying connected to the center.
4	Narrow-web fabric	P2P fact proposals, RAG slices, and LLM context packets traverse the sphere.
5	Human lens	AR/VR inspection, 360-degree movement, and detail cards.

The hardware-facing claim is deliberately conservative: WNW is digitally prepared for future optically addressable molecules or quantum-dot mediated addressing, but today it ships a deterministic schema and browser demo, not a molecule-control product.

3. Quantum Forge

At one end of the WNW spectrum, Gither becomes a shared forge for quantum circuit code. It should feel GitHub-like for collaboration and MLflow-like for experiment lineage, while signed records in the fabric remain the authority.

Surface	Record	Purpose
Circuit repo	QASM, QIR, Qiskit, Cirq, PennyLane, reversible libraries	Code, issues, reviews, signed refs, and merge gates.
Experiment run	backend, shots, seed, transpiler, calibration hash, queue time, result CID	Run lineage without one MLflow server owning truth.
Benchmark frontier	Toffoli, depth, qubits, fidelity, cost, integer score	Reproducible comparison before accepting a claimed improvement.

Example record: `quantum_run = {repo, circuit_cid, language, backend, shots, params_cid, result_cid, score_int, review_sig}`.

4. Vank Mesh Exchange

At the other end of the spectrum, Vank provides a non-custodial P2P market layer. It may borrow familiarity from Bitcoin.de, Intersango, and Mt. Gox-era exchange UX, but it must not recreate their custody concentration. Users keep keys; the fabric gossips signed market facts.

Component	Responsibility	Must not do
Order intent	Signed pair, side, amount, price tick, expiry, jurisdiction hints.	Custody funds.
Gossip orderbook	Merge fresh signed offers from peers and narrow-web relays.	Become canonical truth by server position.
Escrow term	HTLC, multisig, bonded mediator, payment proof, or local rail condition.	Hide settlement rules off-record.
Pulse settlement	Record opened, locked, paid, released, disputed, resolved.	Accept floats on canonical value path.

Example record: `vank_order = {pair, side, amount_int, price_tick, expiry_beat, escrow, proof_cid, signatures}`.

5. Specify Flow

The initial scan returns the proposed fact table before the expensive retrieval. For the example 'all steel mines with detail cards in the world', WNW proposes rows, columns, payload size, and initial entities.

Field	Example estimate
Rows	9,840
Columns	18
Payload	4.6 MB
Relay packets	About 33k x 140-character packets
Initial entities	Carajas Serra Norte, Kiruna, Pilbara iron ore system, Sishen
Decision	Accept, narrow scope, or widen to public web fallback

The proposal record should contain:

- terms and normalized domain
- source scope: nearby, known peers, world, or fallback web
- columns and expected row count
- estimated byte payload and small-device packet count
- initial entity list and unresolved fields
- provenance confidence and Pulse beat

6. Small-Device Relay

For bitchat-like communication, WNW should send the proposal, not the whole table. A BC2-style capsule can carry scope, estimates, beat, and content-address hints in two short lines. The full slice is fetched only after acceptance.

Line	Example
1	WNW1 steel-mines rows=9840 cols=18 bytes=4.6MB beat=0
2	CID? est=<digest> scope=world accept=narrow/refine/query

Security stance: this is field relay and proposal transport, not private-key storage. When secrecy is needed, keep the BC2 convention that each message gets a fresh message wallet.

7. Proposed Backlog

Priority	Item	Definition of done
P0	Specify contract	Signed proposal record with terms, scope, columns, estimates, source layer, and accept/refine decision.
P0	Fact table planner	Deterministic estimate-before-fetch planner with cached estimates per Pulse beat.
P1	BC2 adapter	Two-line proposal capsules for bitchat-style field relay.
P1	Quantum Weft schema	Canonical radial/angular address type with layer, theta, phi, beat, band, and relation digest.
P1	AR/VR fabric view	WebXR demo that inspects shells, traverses 360 degrees, and opens detail cards.
P2	Quantum circuit bridge	Circuit submissions mapped into weft coordinates and compared as signed QPU/GPU/TPU records.
P2	Quantum Forge runs	MLflow-like run records for circuit code: params, backend, shots, result CID, score, and review signature.
P2	Vank mesh orderbook	Non-custodial signed order intents, gossip merge rules, expiry, escrow terms, and settlement proof states.
P2	P2P RAG slices	Bounded fact slices attached to LLM context with provenance and byte estimates.
P3	Molecule addressing research	Track laser, quantum-dot, and molecular-qubit work without claiming hardware readiness too early.
P3	Post-2036 digital twin	Keep today's protocol ready for future molecule fabrics while shipping useful web facts now.

8. Source Anchors

- User-provided ScienceDirect PII: <https://www.sciencedirect.com/science/article/abs/pii/S002223131100723X>. Metadata was not exposed in this environment.
- Nobel Prize in Chemistry 2023 press release on quantum dots: <https://www.nobelprize.org/prizes/chemistry/2023/press-release/>
- RFC 9340, Architectural Principles for a Quantum Internet: <https://www.rfc-editor.org/rfc/rfc9340>
- Bayliss et al., Enhancing Spin Coherence in Optically Addressable Molecular Qubits through Host-Matrix Control: <https://arxiv.org/abs/2204.00168>
- Journal of Luminescence spherical quantum-dot example DOI: <https://doi.org/10.1016/j.jlumin.2022.119181>